Zhuoyang Zou

(773) 290-4562 zhuoyangzou@psu.edu Personal Website zyzou8 Google Scholar LinkedIn

Research Interests

- **Agentic AI for Science and Healthcare.** Building multi-agent LLM systems that automate real-world scientific workflows, including AI-assisted peer review, clinical decision support, and disease prediction.
- **LLM Reasoning and Agent Design.** Optimizing agent architectures for complex reasoning: adaptive planning, multi-agent debate, tool use, and long-horizon task execution.
- **Evaluation and Trustworthy AI.** Designing benchmarks and detection methods that surface when models exploit spurious or stylistic shortcuts rather than genuine reasoning, and building evaluation pipelines that isolate substantive signal from artifact.

Education

Pennsylvania State University *Sep. 2024 – Present*
Ph.D. in Computer Science, Advisor: Prof. Wenpeng Yin *University Park, PA*

Northwestern University *Sep. 2022 – Dec. 2023*
M.S. in Artificial Intelligence *Evanston, IL*

The Chinese University of Hong Kong *Sep. 2017 – May 2022*
B.S. in Computer Science, Advisor: Prof. Qi Dou *Hong Kong*

Honors and Awards

- **ICDS Rising Researcher Award**, Penn State Institute for Computational and Data Sciences (semester stipend and tuition support), 2026
- **Workday AI PhD Fellowship** ($100,000 grant), 2026
- **NSF Research Traineeship Fellow**, LinDiv: Linguistic Diversity Across the Lifespan (#2125865), Pennsylvania State University, 2026 – 2027
- **Workshop Best Paper Award**, ICML 2025, for AAAR-1.0: Assessing AI's Potential to Assist Research

Publications

- **Z. Zou**, A. Ansari, J. Yang, D.C. Zhang, Q. Chen, D. Lee, W. Yin. **Who Wrote It Is Not Enough: Detecting Who Contributed the Insight**. *arXiv:2610.07365*, 2026. [arXiv]
- **Z. Zou**, A. Ansari, D.C. Zhang, D. Lee, W. Yin. **DIAGPaper: Diagnosing Valid and Specific Weaknesses in Scientific Papers via Multi-Agent Reasoning**. *COLM 2026*. [arXiv]
- J. Ahn, R. Kamoi, B. Atil, R. Lou, W. Kang, H. Park, S. Das, **Z. Zou**, et al. **Bridging the Know-Act Gap via Task-Level Autoregressive Reasoning**. *arXiv:2603.22619*, 2026. [arXiv]
- A. Ansari, D.C. Zhang, **Z. Zou**, W. Yin, D. Lee. **M2-Verify: A Large-Scale Multidomain Benchmark for Checking Multimodal Claim Consistency**. *arXiv:2604.01306*, 2026. [arXiv]
- B. Belland, W. Yin, **Z. Zou**, J. Mussoline, C. Kim. **AI-Enabled Automated Scaffolding for Undergraduate Students' Learning to Debug and Reason**. *AI4HE @ CSEDU 2026*. [PDF]
- R. Lou, H. Xu, S. Wang, J. Du, R. Kamoi, X. Lu, J. Xie, Y. Sun, Y. Zhang, J. Ahn, **Z. Zou**, et al. **AAAR-1.0: Assessing AI's Potential to Assist Research**. *ICML 2025 Workshop*, **Best Paper Award**. [arXiv]
- Y. Zhang, W. Zheng, A. Madasu, P. Shi, R. Kamoi, H. Zhou, **Z. Zou**, et al. **HRScene: How Far Are VLMs from Effective High-Resolution Image Understanding?** *ICCV 2025*. [arXiv]
- **Z. Zou**, P.T. Teo, A. Yalamanchili, M. Abazeed. **Optimizing Deep Learning Models: CNN-RNN Augmentation with Grad-CAM Analysis for Predicting Pneumonitis**. *Int. J. Radiation Oncology, Biology, Physics*, 2024. [Paper]

Research Experience

Pennsylvania State University *Sep. 2024 – Present*
Researcher, Agentic AI & Multi-Agent LLM Systems *University Park, PA*

- **InsightProv** *[Qwen2.5-72B, LoRA, adversarial training]*: Insight-provenance detection for scientific reviews: identifies whether a review's insights come from humans, LLMs, or both. Built **InsightProv-v0**, 12,000+ reviews on 4,000+ papers with sentence-level provenance labels; two-stage adversarial framework unlearns stylistic authorship shortcuts. Found AI insights stay close to generic or paper-provided content, while human insights add external knowledge and independent judgment.
- **DIAGPaper** *[Python, GPT-4o, Claude, Llama, LangChain]*: Multi-agent LLM framework for automated scientific peer review. Customizer dynamically generates paper-specific criteria; Rebuttal module runs adversarial author–reviewer debate, filtering 40–60% of invalid critiques; Prioritizer ranks weaknesses by severity, learned from 15,343 real reviews. **F1 51.89 on AAAR, a 9.4% improvement over GPT-4o.**
- **AAAR-1.0** *[Python, NLP pipeline, benchmark design]*: Large-scale benchmark for AI-assisted research. **ICML 2025, Workshop Best Paper Award.**

- **M2-Verify** *[VLMs, multimodal reasoning]*: Multidomain benchmark for multimodal claim consistency across text and image.
- **Bridging the Know-Act Gap** *[LLM agents, autoregressive reasoning]*: Task-level autoregressive reasoning framework closing the gap between LLM knowledge and executable action in agentic settings.

Northwestern University Medical School *Jul. 2023 – May 2024*
Researcher, Medical AI *Chicago, IL*

- **iTox** *[PyTorch, CNN-RNN, Grad-CAM]*: Pneumonitis risk prediction from pre-treatment CT scans. **AUC 0.84.** Grad-CAM localizes predictive lung regions; iTox dose metric reduces pneumonitis risk by **20%.**

The Chinese University of Hong Kong *Jan. 2022 – Aug. 2022*
Researcher, VR & Surgical AI *Hong Kong SAR*

- **VR Surgical Training System** *[Unity C#, stereoscopic rendering]*: Simulated Da Vinci surgical procedures with stereoscopic vision for trainee surgeons.
- **Surgical Instrument Segmentation** *[U-Net, PyTorch]*: Segmentation and activity recognition on surgical video; eye tracking for adaptive training.

Industry Experience

High Fashion Group *May 2024 – Aug. 2024*
AI Specialist *Hong Kong*

- **Fabric Recognition System** *[PyTorch, Siamese Network, ResNet]*: Few-shot image similarity model for fabric pattern matching and quality control.
- **Semantic Fabric Search** *[LLM, RAG, vector embeddings]*: Text-to-fabric retrieval using LLM query encoding and dense retrieval over a product database.
- **Trend Forecasting RAG** *[RAG, LLM, structured data]*: Multi-source pipeline for fashion trend analysis and cost-performance reporting.

Karmo International Company Ltd. *Jun. 2022 – May 2023*
Artificial Intelligence Developer *Hong Kong*

- **Construction Safety Detection** *[YOLOv5/v8, OpenCV]*: Real-time safety gear detection deployed at active construction sites.
- **Autonomous Vehicle Detection** *[TensorFlow, object detection]*: Multi-class real-time detection deployed at Hong Kong Science Park.
- **Drainage Prediction System** *[Django, Random Forest, scikit-learn]*: Web app and predictive model for Hong Kong Government Drainage Services.

Technical Skills

- **LLMs & Agents:** GPT-4o, Claude (Sonnet/Opus), Llama 3, Qwen, Gemini, LangChain, LangGraph, AutoGen, RAG, tool use, chain-of-thought, prompt engineering, multi-agent orchestration
- **AI / ML:** PyTorch, TensorFlow, HuggingFace Transformers, CNN, RNN, YOLO, U-Net, Siamese Networks, Grad-CAM, scikit-learn
- **Data Science & Statistics:** Pandas, NumPy, statistical analysis, hypothesis testing, experiment design, model evaluation & benchmarking, data visualization
- **Dev & Tooling:** Python, Claude Code, Cursor, GitHub Copilot, Docker, Git, Linux, FastAPI, Django, Unity C#
- **Data & Infra:** SQL, FAISS, ChromaDB, AWS, distributed training (DeepSpeed ZeRO-3, multi-GPU), Weights & Biases, Jupyter